





Python Programming Bootcamp – Week 6, Day 02

**Leveraging Python's Ecosystem for Enhanced
Functionality**

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Objectives

- Understand the role of external libraries in Python.
- Explore popular Python packages and libraries.
- Learn how to install, import, and use external libraries in projects.
- Hands-on examples with a few key libraries.



Why Use External Libraries?

- **Expands Python's capabilities** beyond standard libraries.
- **Saves time** by using pre-built, optimized code for complex functions.
- **Supports specific domains:** data science, web development, machine learning, etc.
- Promotes **code reusability** and standardization



What Are Packages and Libraries?

- **Library:** A collection of functions and methods that allows for performing many actions without writing code from scratch (e.g., math, datetime).
- **Package:** A directory containing modules and an `__init__.py` file to be recognized as a package.
- **Examples:** Pandas for data manipulation, Requests for HTTP requests.



Popular Python Libraries

- **For Data Science:** Pandas, NumPy, Matplotlib, Seaborn
- **For Web Development:** Flask, Django, Requests
- **For Machine Learning:** Scikit-learn, TensorFlow, PyTorch
- **For APIs and Automation:** Requests, BeautifulSoup, Selenium



Installation with **pip**

- **pip** is the Python package installer: allows downloading and installing libraries from the Python Package Index (PyPI).
- `pip install <package_name>`
- `pip uninstall <package_name>`
- `pip show <package_name>`
- Example: `pip install requests` installs the **requests** library for making HTTP requests.



Using Installed Libraries

- **Importing a Library:**
 - Use **import** keyword to bring in a library.

```
import requests  
response = requests.get("https://api.github.com")  
print(response.status_code)
```


Using NumPy for Array Operations

```
import numpy as np
arr = np.array([1, 2, 3, 4])
print("Array:", arr)
print("Sum:", np.sum(arr))
```

Data Manipulation with Pandas

Pandas: Essential for data manipulation and analysis, especially with dataframes.

```
import pandas as pd
data = {'Name': ['John', 'Anna'], 'Age': [28, 24]}
df = pd.DataFrame(data)
print(df)
```



Data Visualization with Matplotlib

- **Matplotlib:** Used for creating static, interactive, and animated visualizations in Python.

```
import matplotlib.pyplot as plt  
plt.plot([1, 2, 3], [4, 5, 6])  
plt.show()
```



Working with JSON Data Using Requests

- Requests: Allows easy access to REST APIs, often returning JSON.

```
import requests
path = 'https://jsonplaceholder.typicode.com/posts/1'
response = requests.get(path)
data = response.json()
print(data)
```



Good Practices When Using External Libraries

- Check documentation and compatibility.
- Use virtual environments to manage dependencies.
- Test and lock library versions for consistent functionality.



Hands-On with Requests and JSON

- Install requests library.
- Use requests.get to fetch JSON data from an API.
- Parse and display specific fields in the JSON data.